Summary-statistics Mendelian randomization and GATE score

Methods and application to PDCD1 protein and type 1 diabetes

2026-06-02

Contents

Part I

Theoretical derivations

1 Recovering multivariate regression coefficients from univariate summary statistics

1.1 Introduction

In genome-wide association studies (GWAS), summary statistics are typically reported as *marginal* (univariate) regression coefficients $\hat{\beta}_{u,j}$, one per SNP, estimated by regressing the trait on each SNP separately. When a locus contains P SNPs in linkage disequilibrium (LD), the marginal coefficient for SNP j conflates its direct effect on the trait with indirect contributions absorbed from correlated neighbours.

This section derives the formula for recovering *multivariate* (joint) regression coefficients $\hat{\boldsymbol{\beta}}_m$ from the marginal coefficients, using the LD correlation matrix among SNPs. The derivation works with **unstandardized** genotype units (allele dosage 0/1/2) throughout: both the input ($\hat{\boldsymbol{\beta}}_u$) and the output ($\hat{\boldsymbol{\beta}}_m$) are in units of trait change per additional copy of the effect allele.

1.2 Notation

Let n be the number of individuals and P the number of SNPs in a locus.

- $\sigma_j^2 = \mathrm{Var}(g_j) = 2f_j(1-f_j)$: genotype variance for SNP j.
- $\mathbf{D} = \mathrm{diag}(\sigma_1, \ldots, \sigma_P)$: diagonal matrix of SDs.
- $\mathbf{G}^* = \mathbf{G}\mathbf{D}^{-1}$: genotype matrix scaled to unit variance per column.
- $\mathbf{C}_g = \mathbf{G}^{*\top}\mathbf{G}^*/n$: $P \times P$ LD correlation matrix.
- $\boldsymbol{\Sigma}_g = \mathbf{G}^\top\mathbf{G}/n = \mathbf{D}\mathbf{C}_g\mathbf{D}$: genotype covariance matrix.

1.3 Key identity

The marginal-to-multivariate relationship follows from the normal equations $\boldsymbol{\Sigma}_g\hat{\boldsymbol{\beta}}_m = (1/n)\mathbf{G}^\top\mathbf{y} = \mathbf{D}^2\hat{\boldsymbol{\beta}}_u$:

$$\boxed{\hat{\boldsymbol{\beta}}_m = \mathbf{D}^{-1}\,\mathbf{C}_g^{-1}\,\mathbf{D}\,\hat{\boldsymbol{\beta}}_u.} \tag{1}$$

Both $\hat{\boldsymbol{\beta}}_u$ and $\hat{\boldsymbol{\beta}}_m$ are in **units of trait change per allele**; $\mathbf{D}$ converts between per-allele and per-SD units internally. Equivalently: scale marginal coefficients to per-SD units, apply $\mathbf{C}_g^{-1}$, rescale back.

Verification: $\boldsymbol{\Sigma}_g\hat{\boldsymbol{\beta}}_m = \mathbf{D}\mathbf{C}_g\mathbf{D}\cdot\mathbf{D}^{-1}\mathbf{C}_g^{-1}\mathbf{D}\hat{\boldsymbol{\beta}}_u = \mathbf{D}^2\hat{\boldsymbol{\beta}}_u = (1/n)\mathbf{G}^\top\mathbf{y}$. ✓

1.4 Application to GWAS summary statistics

In practice $\hat{\mathbf{C}}_g$ is estimated from a reference panel. When $P \gg n_{\mathrm{ref}}$ or SNPs are in near-perfect LD, the **truncated pseudoinverse** is used: retain only the K eigenvalues $d_k > \epsilon_0 d_1$ (default $\epsilon_0 = 0.01$):

$$\hat{\boldsymbol{\beta}}_m = \hat{\mathbf{D}}^{-1}\,\hat{\mathbf{C}}_g^{+}\,\hat{\mathbf{D}}\,\hat{\boldsymbol{\beta}}_u. \tag{2}$$

The effective rank K is reported for each locus.

1.5 Scalar instrument construction

The P per-allele multivariate coefficients are combined into a scalar instrument:

$$Z_i = \frac{S_i}{\hat{\alpha}}, \qquad S_i = \mathbf{g}_i^\top\hat{\boldsymbol{\beta}}_m, \qquad \hat{\alpha} = \|\hat{\boldsymbol{\beta}}_m\|.$$

The regression coefficient of exposure X on Z equals $\hat{\alpha}$, confirming $\mathbb{E}[X \mid Z] \approx \hat{\alpha} Z$.

1.6 Standard error of $\hat{\alpha}$

The Fisher information from N observations of $X = \alpha Z + \varepsilon_X$ is $\mathcal{I}_\alpha = N\ \mathrm{Var}(Z)/(\sigma_X^2 - \mathrm{Var}(S))$, where σ_X^2 is estimated from per-SNP marginal SEs via $\sigma_X^2 \approx \sigma_j^2 (N\,\mathrm{SE}(\hat{\beta}_{u,j})^2 + \hat{\beta}_{u,j}^2)$ (median over j). The SE is

$$\boxed{\mathrm{SE}(\hat{\alpha}) = \sqrt{\frac{\sigma_X^2 - \widehat{\mathrm{Var}}(S)}{N\,\widehat{\mathrm{Var}}(Z)}}.} \tag{3}$$

2 Instrument-outcome coefficient $\hat{\gamma}$

2.1 Multivariate outcome coefficients

The per-allele univariate SNP-outcome coefficients $\hat{\boldsymbol{\gamma}}_u$ are recovered from z-scores as $\hat{\gamma}_{u,j} = z_j / \sqrt{N_Y \sigma_j^2 p(1-p)}$. The same LD inversion gives the multivariate outcome coefficients:

$$\hat{\boldsymbol{\gamma}}_m = \hat{\mathbf{D}}^{-1}\,\hat{\mathbf{C}}_g^{+}\,\hat{\mathbf{D}}\,\hat{\boldsymbol{\gamma}}_u, \tag{4}$$

using the same $\hat{\mathbf{D}}$, $\hat{\mathbf{C}}_g^{+}$ as for the exposure.

2.2 Closed-form estimator

Projecting the outcome score $T_i = \mathbf{g}_i^\top \hat{\boldsymbol{\gamma}}_m$ onto the scalar instrument Z gives

$$\hat{\gamma} = \hat{\alpha}\,\frac{\hat{\boldsymbol{\gamma}}_m^\top \hat{\boldsymbol{\Sigma}}_g \hat{\boldsymbol{\beta}}_m}{\hat{\boldsymbol{\beta}}_m^\top \hat{\boldsymbol{\Sigma}}_g \hat{\boldsymbol{\beta}}_m} = \hat{\alpha}\,\frac{\sum_{j=1}^{P} \hat{\sigma}_j^2\,\hat{\gamma}_{u,j}\,\hat{\beta}_{m,j}}{\widehat{\mathrm{Var}}(S)}. \tag{5}$$

$\hat{\gamma}$ is a weighted sum of the per-allele products $\hat{\gamma}_{u,j}\hat{\beta}_{m,j}$ with weights proportional to genotype variance $\hat{\sigma}_j^2$.

2.3 Standard error of $\hat{\gamma}$

For a binary outcome (proportion of cases p):

$$\boxed{\mathrm{SE}(\hat{\gamma}) = \frac{1}{\sqrt{N_Y\,\widehat{\mathrm{Var}}(Z)\,p(1-p)}} = \frac{\hat{\alpha}}{\sqrt{N_Y\,\widehat{\mathrm{Var}}(S)\,p(1-p)}}.} \tag{6}$$

Here $\widehat{\mathrm{Var}}(Z) = \widehat{\mathrm{Var}}(S)/\hat{\alpha}^2$ is computed from the reference panel.

3 GATE score coefficient from summary-level pQTL statistics

3.1 The GATE score

The GATE (Genetically Aided Trait Estimation) score sums genetically predicted exposure values over all trans-pQTLs:

$$G_i = \sum_{k \in \mathrm{trans}} \hat{\alpha}_k\,Z_{ik}, \qquad G_i^* = G_i \,/\, \mathrm{SD}_{\mathrm{ref}}(G),$$

where $\mathrm{SD}_{\mathrm{ref}}(G) = \sqrt{\sum_k \hat{\alpha}_k^2\,\mathrm{sd}_{Z,k}^2}$ (trans-loci are independent so cross-terms vanish).

3.2 Score test for $\hat{\beta}_{\mathrm{GATE}}$

Under the null logistic model with residuals $r_i = Y_i - \hat{\mu}_i^{(0)}$ and weights $w_i = \hat{\mu}_i^{(0)}(1 - \hat{\mu}_i^{(0)})$, the composite score and partial information are (using independence across loci):

$$U_{G^*} = \frac{1}{\mathrm{SD}_G} \sum_k \hat{\alpha}_k\, U_k, \qquad V_{G^*} = \frac{1}{\mathrm{SD}_G^2} \sum_k \hat{\alpha}_k^2\, V_k.$$

Substituting $U_k = \hat{\gamma}_k V_k$ (score estimate for locus k) and $V_k = 1/\mathrm{SE}(\hat{\gamma}_k)^2$:

$$\boxed{\hat{\beta}_{\mathrm{GATE}} = \mathrm{SD}_G \cdot \frac{\displaystyle\sum_k \hat{\alpha}_k\, \hat{\gamma}_k \,/\, \mathrm{SE}(\hat{\gamma}_k)^2}{\displaystyle\sum_k \hat{\alpha}_k^2 \,/\, \mathrm{SE}(\hat{\gamma}_k)^2}, \qquad \mathrm{SE}(\hat{\beta}_{\mathrm{GATE}}) = \frac{\mathrm{SD}_G}{\sqrt{\displaystyle\sum_k \hat{\alpha}_k^2 \,/\, \mathrm{SE}(\hat{\gamma}_k)^2}}.} \tag{7}$$

3.3 Simplification of the SE

Substituting $V_k = N\, \mathrm{sd}_{Z,k}^2\, p(1-p)$ from eq. (6) gives $\sum_k \hat{\alpha}_k^2 V_k = Np(1-p)\,\mathrm{SD}_G^2$, so:

$$\mathrm{SE}(\hat{\beta}_{\mathrm{GATE}}) = \frac{1}{\sqrt{N\, p(1-p)}}. \tag{8}$$

The SE equals that of a single 1-df binary logistic regression. The z-statistic simplifies to the IVW z-statistic with weights $w_k = (\hat{\alpha}_k^*)^2$ (where $\hat{\alpha}_k^* = \hat{\alpha}_k\, \mathrm{sd}_{Z,k}$ are the per-SD(Z) exposure coefficients):

$$z_{\mathrm{GATE}} = \sqrt{N\, p(1-p)} \cdot \frac{\sum_k (\hat{\alpha}_k^*)^2\, \hat{\theta}_k}{\sqrt{\sum_k (\hat{\alpha}_k^*)^2}}, \qquad \hat{\theta}_k = \hat{\gamma}_k^* / \hat{\alpha}_k^*. \tag{9}$$

3.4 Comparison: summary-level vs direct regression (PDCD1/T1D)

Table 1: GATE coefficient for T1D ($\hat{\beta}_{\mathrm{GATE}}$ = log-OR per SD of GATE score) from summary-level formula (eq. 7) vs direct individual-level regression. Trans-pQTLs: loci contributing to the GATE score (PDCD1 cis-pQTL excluded). SE (summary-level) = $1/\sqrt{Np(1-p)} \approx 0.0184$ (eq. 8).

Method	$\hat{\beta}$	SE	z	p	Trans pQTLs
Summary-level	0.2244	0.01837	12.218	2.5e-34	38
Direct (individual-level)	0.1980	0.02044	9.687	3.42e-22	41

Part II

PDCD1/T1D MR-Hevo analysis

4 Introduction

We estimate the causal effect of circulating PDCD1 (programmed cell death protein 1) levels on risk of type 1 diabetes (T1D) using Mendelian randomization with the regularized horseshoe prior (MR-Hevo).

The T1D outcome dataset is the same across all three analyses: cases from the Scottish T1D Bioresource (SDRNT1BIO) and controls from Generation Scotland.

Results are presented from three analyses, differing in the LD reference panel used to construct scalar instruments and in whether instrument-outcome coefficients $\hat{\gamma}$ are estimated from individual-level genotypes or from GWAS summary statistics:

1. **Analysis 1 (Zhou et al.)**: $\hat{\alpha}$ and $\hat{\gamma}$ from Zhou et al., who used the 1000 Genomes EUR panel (503 samples) as the LD reference.
2. **Analysis 2 (directly computed $\hat{\gamma}$, UKBB 18M EUR LD)**: $\hat{\alpha}$ from UK Biobank Olink proteomics ($N = 33{,}902$); $\hat{\gamma}$ computed by scoring each individual on $Z_i = \sum_j G_{ij}\,\hat{\beta}_{m,j}$ and regressing T1D on Z_i (4,922 cases, 7,452 controls; adjusted for sex, age, and the first three principal components). Instrument weights $\hat{\beta}_m$ from the UKBB EUR LD matrix (362,063 samples, 18 million variants).
3. **Analysis 3 (LD approximate $\hat{\gamma}$, UKBB 18M EUR LD)**: same $\hat{\alpha}$ and instrument weights $\hat{\beta}_m$ as Analysis 2; $\hat{\gamma}$ estimated from per-SNP SNP-outcome coefficients via the LD approximate projection formula (Section~2), using the same UKBB EUR LD matrix.

Scalar genetic instruments are constructed from loci associated with PDCD1 at $p < 10^{-6}$, excluding the HLA region (chromosome 6: 25–34 Mb), with minor allele frequency ≥ 0.01. All displayed values of $\hat{\alpha}$ and $\hat{\gamma}$ are in per-SD(Z) units.

5 Data sources

Table 2: Source data files used in this analysis.

File	Description	Source
`PDCD1_Q15116_OID21396_v1_Oncology.tar`	SNP–protein GWAS summary statistics for PDCD1 (UniProt Q15116; Olink Oncology panel, assay OID21396); $N = 33{,}902$ UK Biobank EUR participants	UK Biobank Pharma Proteomics Project (UKB-PPP); Sun et al. (2023) *Nature* 622:329
`z\_scores\_gs.rds`	Per-SNP T1D SNP-outcome z-statistics computed from individual-level SDRNT1BIO/GS genotypes by score test; 4,922 cases, 7,452 controls; aligned to `bim_a1` allele convention	Derived from GS plink files; output of `compute_gamma_indiv.R`
`pdcd1.mr.coeffs.csv`	Scalar $\hat{\alpha}$ and $\hat{\gamma}$ coefficients from Analysis 1 (Zhou et al.), with 1000 Genomes EUR LD	Computed by Zhou et al. from UKB-PPP Olink data and SDRNT1BIO/GS individual-level data
`LD_Eur_18mvariants/int8/` (genoscores)	Block-LD matrix; UKBB EUR participants; $N = 362{,}063$; $\sim$18M variants; GRCh37; int8	UK Biobank; computed using magenpy
`GS_T1biochrall_filtered_rsmapped.bed/.bim/.fam` (diabepi)	Individual-level T1D genotypes (SDRNT1BIO cases + GS controls); 25,744 individuals; 7.8M SNPs; GRCh37; plink BED format	Scottish Diabetes Research Network Type 1 Bioresource (SDRNT1BIO) and Generation Scotland (GS)

Table 2: Source data files used in this analysis. *(continued)*

File	Description	Source
`PHENO_T1DM\_...NORELPCA.samples` (diabepi)	T1D case-control phenotype file; 4,922 cases, 7,452 controls; includes sex, age, PC1–3 covariates; Oxford .samples format	SDRNT1BIO/GS; quality-controlled for relatedness, non-T1D diabetes, and population structure
`refpop/kg.2020.hg38.eur.bim`	Variant index (rsid, chr, pos hg38) for 1000 Genomes EUR panel; 503 samples; used for (chr, pos) → rsid lookup	1000 Genomes Project phase 3, GRCh38 liftover
`refGene.txt.gz`	RefSeq protein-coding gene annotations (NM_ accessions); used to label loci with nearest gene	UCSC Genome Browser, hg38 assembly

6 Analysis 1 (Zhou et al.): individual-level $\hat{\gamma}$, 1000 Genomes EUR LD

38 scalar instruments; $\hat{\alpha}$ and $\hat{\gamma}$ from Zhou et al., computed from individual-level UK Biobank Olink proteomics data and SDRNT1BIO/GS T1D genotype–phenotype data, using the 1000 Genomes EUR panel for LD.

Table 3: Instrument coefficients — Analysis 1 (Zhou et al.): 1000 Genomes EUR LD. Nearest Gene: protein-coding gene(s) nearest to or overlapping the locus. Num SNPs: SNPs in locus; $\hat{\alpha}$ and $\hat{\gamma}$: instrument-exposure and instrument-outcome coefficients per SD of the scalar instrument (SE in parentheses). *: $|\hat{\gamma}/\text{SE}(\hat{\gamma})| > 2.58$ (two-sided $p < 0.01$).

Nearest Gene	Chr	Pos (Mb)	Width (Mb)	Num SNPs	$\hat{\alpha}$ (SE)	$\hat{\gamma}$ (SE)	*
ACAP3	1	1.21	0.16	99	0.0253 (0.0051)	0.0188 (0.0206)	
ARTN	1	43.48	0.64	420	0.0540 (0.0051)	0.0220 (0.0201)	
PTPN22	1	113.53	0.60	234	0.0293 (0.0051)	0.1635 (0.0197)	*
HDGF	1	156.77	0.07	71	0.0289 (0.0051)	0.0127 (0.0202)	
AXDND1	1	179.31	0.49	30	0.0357 (0.0051)	-0.0201 (0.0204)	
NEK7	1	198.30	0.16	33	0.0252 (0.0051)	-0.0191 (0.0202)	
C1orf147	1	206.50	0.02	2	0.0257 (0.0051)	-0.0087 (0.0201)	
RN7SL51P	2	62.26	0.12	61	0.0406 (0.0051)	0.0181 (0.0207)	
GLS	2	190.64	0.47	59	0.0363 (0.0051)	0.0262 (0.0199)	
CTLA4	2	203.83	0.10	129	0.0400 (0.0051)	0.1254 (0.0202)	*
PDCD1	2	241.00	0.23	24	0.1237 (0.0051)	-0.0130 (0.0203)	
ARHGEF3	3	56.90	0.02	2	0.0266 (0.0051)	-0.0194 (0.0203)	
GPR160	3	170.00	0.20	132	0.0339 (0.0051)	0.0134 (0.0203)	
ST6GAL1	3	187.02	0.00	6	0.0266 (0.0051)	-0.0325 (0.0201)	
CAPSL	5	35.80	0.12	42	0.0310 (0.0051)	0.0394 (0.0203)	
CDC25C	5	138.25	0.04	3	0.0255 (0.0051)	-0.0071 (0.0199)	
	6	51.28	0.02	9	0.0327 (0.0051)	-0.0414 (0.0202)	
BACH2	6	90.27	0.03	2	0.0265 (0.0051)	0.1428 (0.0199)	*
TAGAP-AS1	6	159.05	0.04	20	0.0271 (0.0051)	0.0617 (0.0203)	*
CD274	9	5.48	0.02	25	0.0277 (0.0051)	0.0250 (0.0200)	
B3GALT9	9	120.75	0.22	138	0.0308 (0.0051)	0.0589 (0.0203)	*
IL15RA	10	5.69	0.38	63	0.0391 (0.0051)	0.0715 (0.0201)	*
ATP5MC1P7	10	69.38	0.16	183	0.0538 (0.0051)	0.0039 (0.0202)	
DYDC1	10	80.28	0.24	225	0.0291 (0.0051)	-0.0178 (0.0201)	
SH2B3	12	109.92	2.88	1186	0.0714 (0.0051)	0.1157 (0.0203)	*
ACADS	12	120.70	0.25	249	0.0328 (0.0051)	0.0239 (0.0200)	
ALDH1A2	15	58.46	0.04	10	0.0402 (0.0051)	0.0124 (0.0200)	
DRAIC	15	69.69	0.07	39	0.0285 (0.0051)	0.0465 (0.0201)	
PEAK1	15	77.15	0.11	6	0.0248 (0.0051)	0.0249 (0.0201)	
CLEC16A	16	11.02	0.09	26	0.0260 (0.0051)	0.1642 (0.0207)	*
ACADVL	17	7.16	0.20	38	0.0473 (0.0051)	0.0288 (0.0200)	
FLJ40194	17	49.24	0.16	106	0.0348 (0.0051)	0.0255 (0.0202)	
CEP131	17	81.17	0.13	147	0.0468 (0.0051)	0.0205 (0.0201)	
DAPK3	19	3.95	0.16	73	0.0371 (0.0051)	0.0232 (0.0202)	
AP1M2	19	10.53	0.55	275	0.0481 (0.0051)	-0.0199 (0.0203)	
FUT2	19	48.65	0.10	105	0.0321 (0.0051)	0.1395 (0.0202)	*
AIRE	21	44.29	0.00	5	0.0256 (0.0051)	-0.0022 (0.0201)	
OGFRP1	22	42.19	0.10	28	0.0255 (0.0051)	-0.0181 (0.0202)	

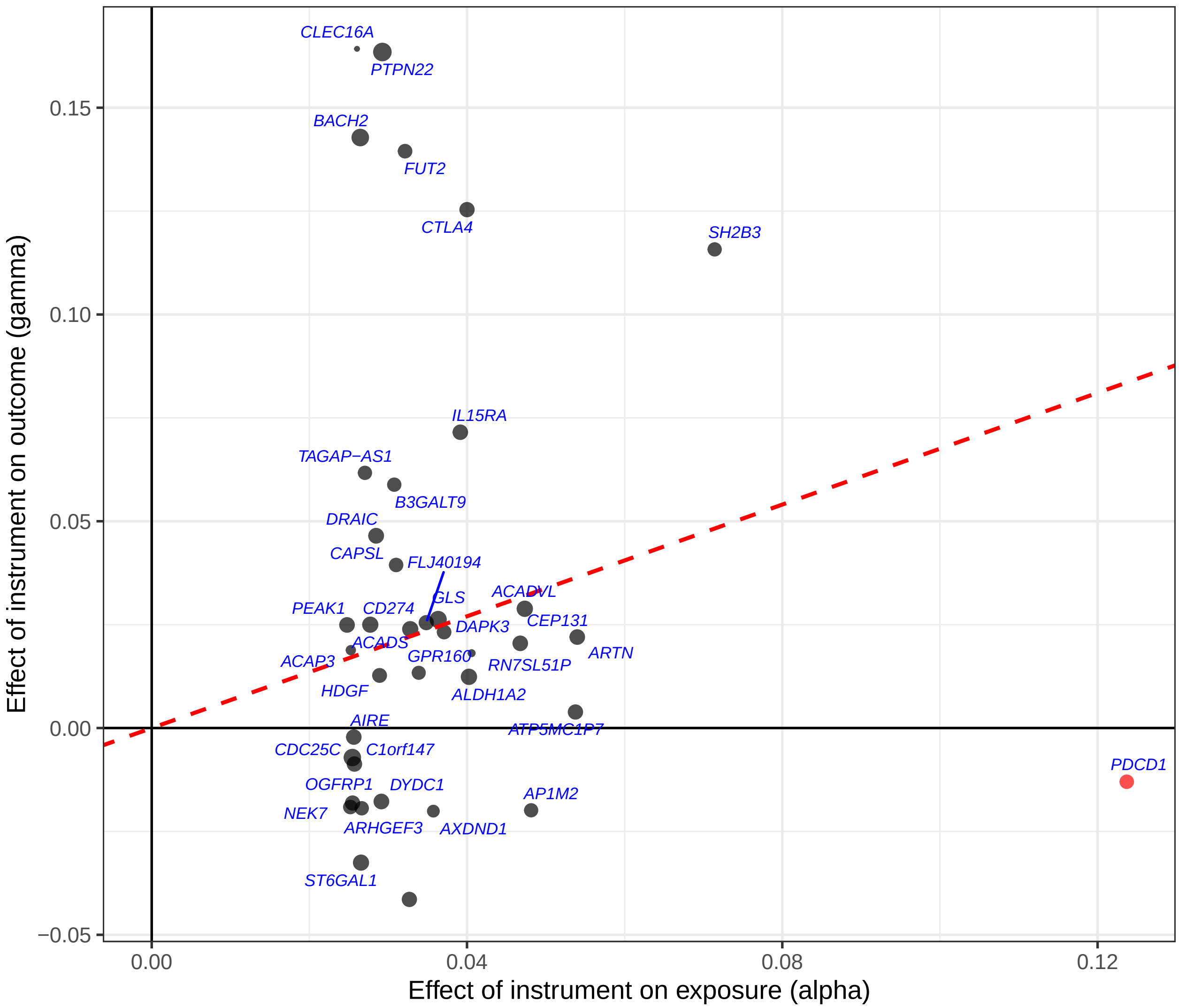

Figure 1: Analysis 1 (Zhou et al.), 1000 Genomes EUR LD. $\hat{\gamma}$ vs $\hat{\alpha}$ (per SD of scalar instrument) for 38 instruments. IVW slope estimated from trans-pQTLs only: $\hat{\theta} = 0.675$ (SE 0.098, $p = 5 \times 10^{-10}$). Red dashed line: IVW slope (trans only). Red point: PDCD1 cis-pQTL.

7 Analysis 2: individual-level $\hat{\gamma}$ from SDRNT1BIO/GS genotypes, UKBB 18M EUR LD

39 scalar instruments; $\hat{\alpha}$ from UK Biobank Olink proteomics ($N = 33{,}902$). $\hat{\gamma}$ estimated from individual-level SDRNT1BIO/GS genotypes (4,922 cases, 7,452 controls), adjusted for sex, age, and the first three genetic principal components. Instrument weights $\hat{\beta}_m$ derived from the UKBB EUR LD matrix (362,063 samples, 18 million variants).

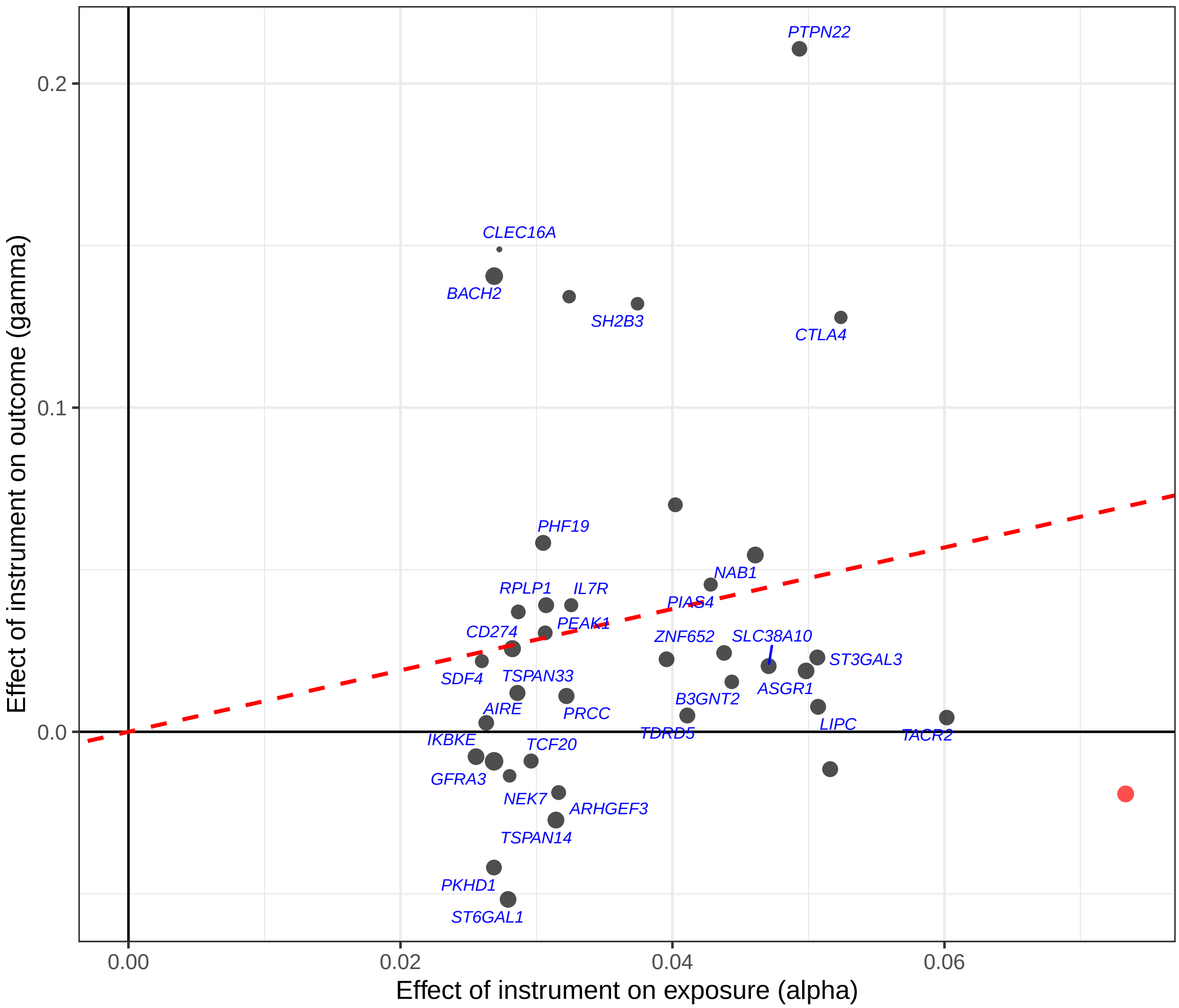

Figure 2: Analysis 2: directly computed $\hat{\gamma}$ (individual-level logistic regression, UKBB 18M EUR LD instrument weights) vs $\hat{\alpha}$ (per SD of scalar instrument). IVW slope estimated from trans-pQTLs only: $\hat{\theta} = 0.948$ (SE 0.088, $p = 4 \times 10^{-30}$). Red dashed line: IVW slope (trans only). Red point: PDCD1 cis-pQTL.

8 Analysis 3: LD approximate $\hat{\gamma}$ from SNP-outcome coefficients, UKBB 18M EUR LD

39 scalar instruments; $\hat{\alpha}$ from UK Biobank Olink proteomics ($N = 33{,}902$). $\hat{\gamma}$ estimated from per-SNP T1D SNP-outcome coefficients (from SDRNT1BIO/GS individual-level logistic regression, 4,922 cases, 7,452 controls), LD-adjusted using the UKBB EUR LD matrix (362,063 samples, 18 million variants). Same instrument weights $\hat{\beta}_m$ as Analysis 2.

Table 4: Instrument coefficients — Analysis 3: LD approximate SNP-outcome coefficients, UKBB 18M EUR LD. Nearest Gene: protein-coding gene(s) nearest to or overlapping the locus. Num SNPs: SNPs in locus; eff rank: effective rank of LD matrix. SD(Z): standard deviation of scalar instrument in reference panel. LD blks: number of zarr LD blocks spanned by the locus. Min MAF: minimum minor allele frequency across SNPs contributing to the locus (UKBB proteomics GWAS). $\hat{\alpha}$ and $\hat{\gamma}$: instrument-exposure and instrument-outcome coefficients per SD of the scalar instrument (SE in parentheses). *: $|\hat{\gamma}/\mathrm{SE}(\hat{\gamma})| > 2.58$ (two-sided $p < 0.01$).

Nearest Gene	Chr	Pos (Mb)	Width (Mb)	Num SNPs	eff rank	SD(Z)	LD blks	Min MAF	$\hat{\alpha}$ (SE)	$\hat{\gamma}$ (SE)
SDF4	1	1.21	0.04	84	2	2.775	1	0.07	0.0260 (0.0051)	0.0197 (0.0184)
ST3GAL3	1	43.33	0.79	353	9	3.149	1	0.06	0.0507 (0.0051)	0.0212 (0.0184)
PTPN22	1	113.53	0.64	191	8	0.580	1	0.01	0.0493 (0.0051)	0.0764 (0.0074)
PRCC	1	156.77	0.07	56	6	0.690	1	0.06	0.0322 (0.0051)	0.0021 (0.0036)
TDRD5	1	179.31	0.49	26	11	0.360	1	0.02	0.0411 (0.0052)	0.0050 (0.0184)
NEK7	1	198.30	0.16	29	2	0.878	1	0.08	0.0280 (0.0052)	-0.0124 (0.0184
IKBKE	1	206.50	0.00	1	1	0.593	1	0.23	0.0256 (0.0051)	-0.0070 (0.0184
B3GNT2	2	62.26	0.12	45	11	0.460	1	0.02	0.0444 (0.0053)	0.0142 (0.0184)
NAB1	2	190.64	0.47	45	7	0.653	1	0.07	0.0461 (0.0052)	0.0502 (0.0184)
CTLA4	2	203.83	0.10	102	7	1.218	1	0.25	0.0524 (0.0052)	0.0582 (0.0092)
PDCD1	2	241.04	0.20	21	15	0.309	2	0.01	0.0733 (0.0053)	-0.0177 (0.0150
ARHGEF3	3	56.90	0.02	2	2	0.579	1	0.08	0.0316 (0.0052)	-0.0170 (0.0184
PHC3, GPR160	3	170.00	0.20	112	6	1.980	1	0.25	0.0396 (0.0052)	0.0209 (0.0184)
ST6GAL1	3	187.02	0.00	6	3	0.836	1	0.31	0.0279 (0.0052)	-0.0474 (0.0184
IL7R	5	35.80	0.12	35	5	2.088	1	0.26	0.0326 (0.0052)	0.0355 (0.0184)
GFRA3	5	138.16	0.09	5	4	0.262	1	0.02	0.0269 (0.0052)	-0.0077 (0.0169
PKHD1	6	51.28	0.02	5	3	1.201	1	0.40	0.0269 (0.0052)	-0.0393 (0.0184
BACH2	6	90.27	0.03	2	2	0.494	1	0.18	0.0269 (0.0051)	0.1287 (0.0184)
	6	159.05	0.04	20	5	0.754	1	0.32	0.0287 (0.0051)	0.0335 (0.0184)
TSPAN33	7	129.10	0.04	4	2	0.978	1	0.31	0.0286 (0.0051)	0.0111 (0.0184)
CD274	9	5.48	0.02	16	2	1.738	1	0.23	0.0282 (0.0051)	0.0236 (0.0184)
PHF19	9	120.75	0.22	108	3	4.386	1	0.26	0.0305 (0.0051)	0.0531 (0.0184)
IL15RA	10	5.69	0.38	46	5	0.906	2	0.12	0.0402 (0.0051)	0.0638 (0.0184)
TACR2	10	69.38	0.16	148	15	1.376	1	0.02	0.0602 (0.0052)	0.0041 (0.0184)
TSPAN14	10	80.28	0.24	180	6	2.740	1	0.15	0.0314 (0.0051)	-0.0249 (0.0184
SH2B3	12	110.00	2.79	993	15	3.069	1	0.03	0.0736 (0.0051)	0.1197 (0.0182)
SPPL3	12	120.75	0.19	206	6	2.523	1	0.28	0.0374 (0.0051)	0.0103 (0.0185)
LIPC	15	58.46	0.06	16	7	0.670	1	0.04	0.0507 (0.0052)	0.0073 (0.0300)
RPLP1	15	69.69	0.07	32	4	1.216	1	0.37	0.0307 (0.0051)	0.0356 (0.0184)
PEAK1	15	77.15	0.09	5	4	0.562	1	0.33	0.0306 (0.0052)	0.0278 (0.0184)
CLEC16A	16	11.04	0.08	19	5	1.346	1	0.23	0.0273 (0.0051)	0.1343 (0.0184)
GABARAP	17	7.16	0.20	34	16	0.670	1	0.08	0.0498 (0.0052)	0.0178 (0.0189)
ZNF652	17	48.49	0.91	109	9	1.361	1	0.12	0.0438 (0.0051)	0.0214 (0.0179)
SLC38A10	17	81.17	0.13	127	10	1.821	1	0.20	0.0471 (0.0051)	0.0194 (0.0188)
PIAS4	19	3.95	0.14	59	13	0.765	1	0.03	0.0428 (0.0052)	0.0422 (0.0184)
DNM2, SMARCA4, KRI1, SLC44A2, C19orf38, CDKN2D, ATG4D, AP1M2, ILF3, YIPF2, CARM1, QTRT1, TMED1, TIMM29	19	10.53	0.55	221	12	2.066	1	0.05	0.0516 (0.0051)	-0.0104 (0.0182
FUT2	19	48.65	0.10	93	5	3.682	1	0.40	0.0324 (0.0052)	0.1228 (0.0184)
AIRE	21	44.29	0.00	3	1	0.764	1	0.11	0.0263 (0.0051)	0.0025 (0.0184)
TCF20	22	42.19	0.10	19	3	0.877	1	0.20	0.0296 (0.0051)	-0.0082 (0.0184

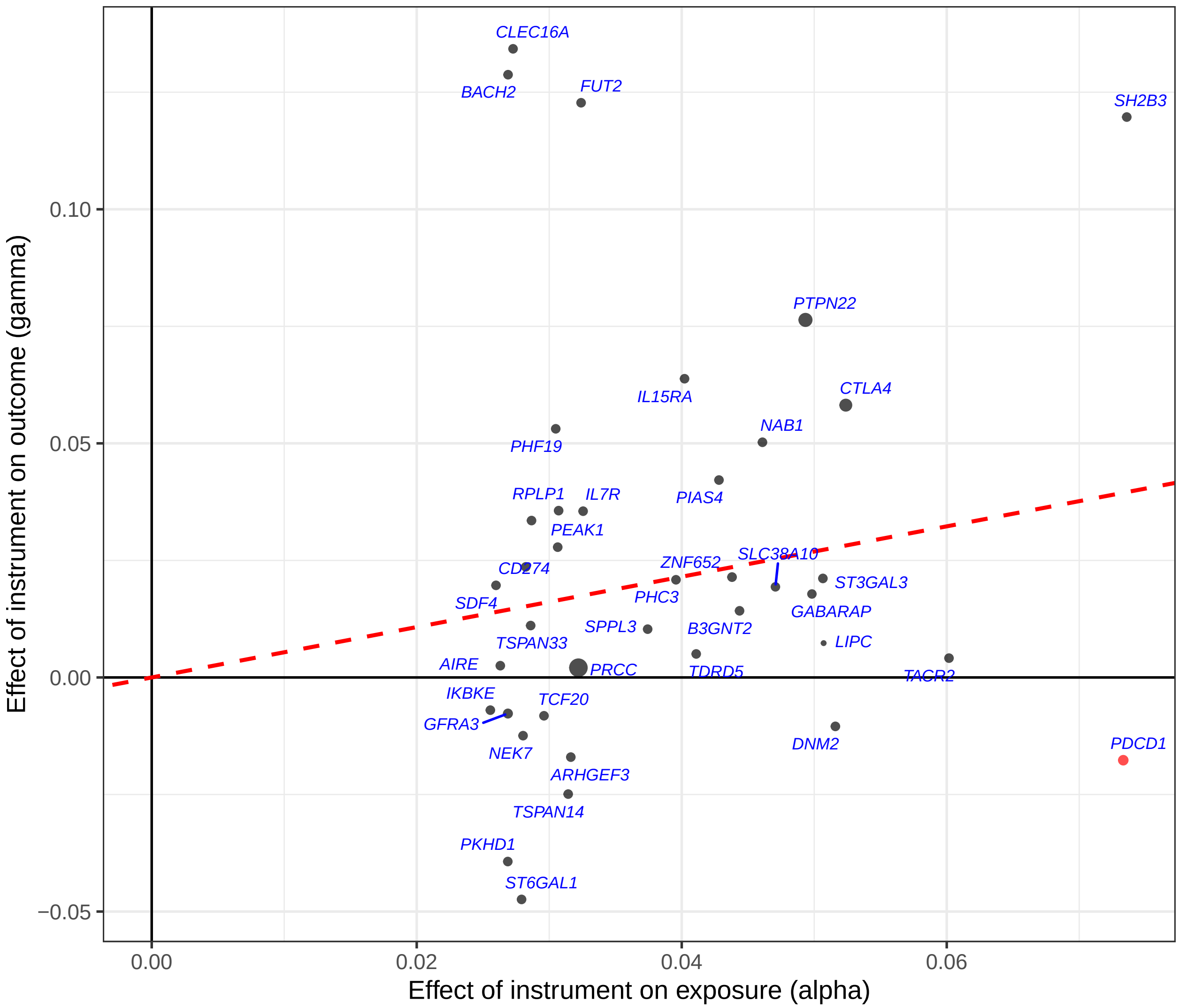

Figure 3: Analysis 3: LD approximate $\hat{\gamma}$ (SNP-outcome coefficients + UKBB 18M EUR LD projection formula) vs $\hat{\alpha}$ (per SD of scalar instrument) for 39 instruments. IVW slope estimated from trans-pQTLs only: $\hat{\theta} = 0.538$ (SE 0.062, $p = 3 \times 10^{-20}$). Red dashed line: IVW slope (trans only). Red point: PDCD1 cis-pQTL.

9 Comparison of $\hat{\alpha}$: Analysis 1 (1000G LD) vs Analyses 2 and 3 (UKBB 18M LD)

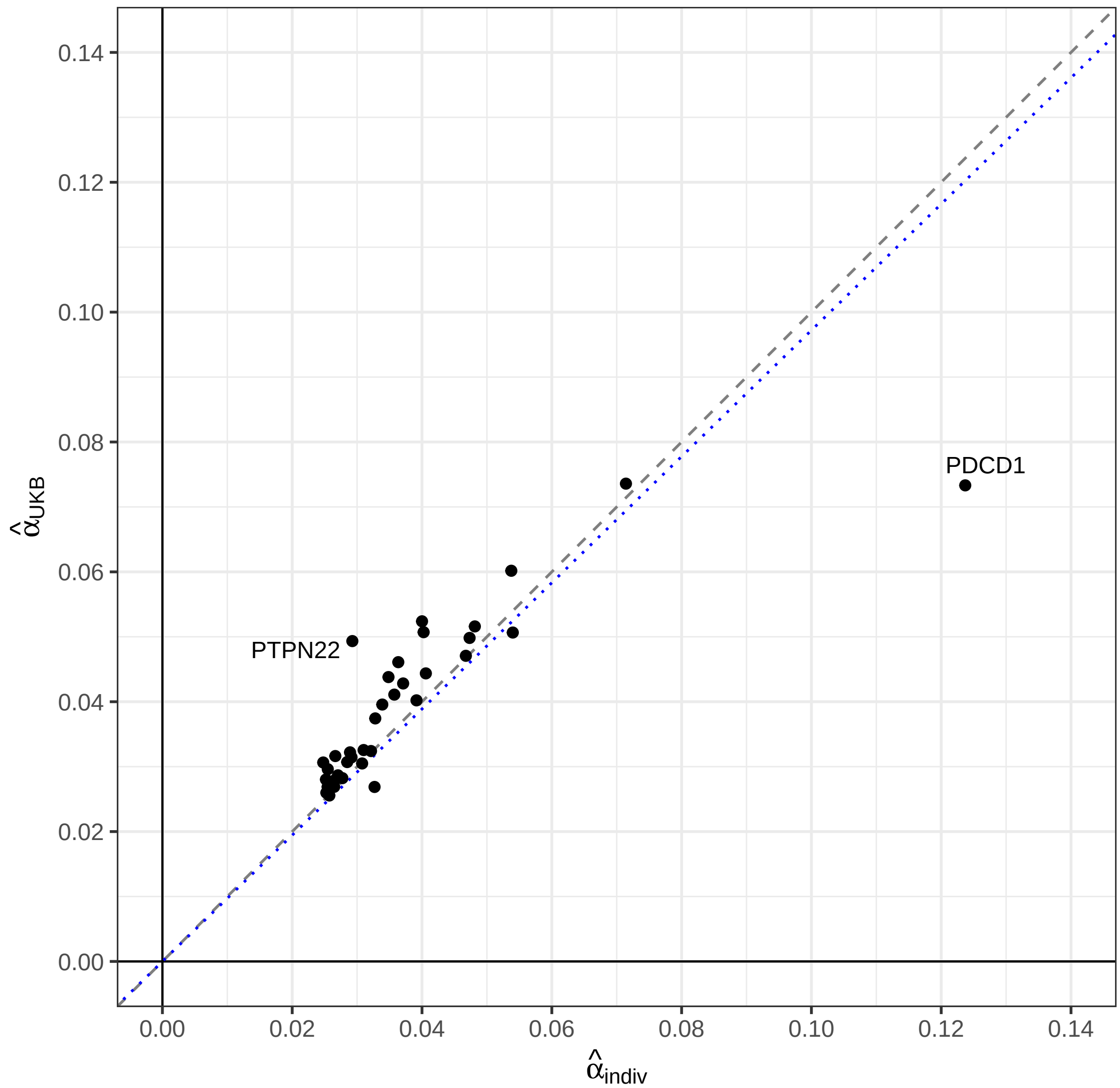

Figure 4: Instrument-exposure coefficient $\hat{\alpha}$ (per SD of scalar instrument): Analysis 1 / Zhou et al. with 1000 Genomes EUR LD (x-axis) vs Analysis 2 with UKBB 18M EUR LD (y-axis). Loci matched by chromosomal overlap. Labelled points are outliers from the identity line ($|y - x| > 2\,\text{SD}$). Grey dashed line: identity ($y = x$). Blue dotted line: OLS regression through the origin.

10 Comparison of directly computed $\hat{\gamma}$: Zhou et al. (1000G LD) vs Analysis 2 (UKBB 18M LD)

Both analyses use individual-level SDRNT1BIO/GS genotypes to compute the scalar instrument score $Z_i = \sum_j G_{ij}\,\hat{\beta}_{m,j}$ for each individual and regress T1D on Z_i. They differ only in the LD reference panel used to derive the instrument weights $\hat{\beta}_m$: 1000 Genomes EUR (503 samples, Analysis 1) vs UKBB EUR 18M variant LD matrix (362,063 samples, Analysis 2).

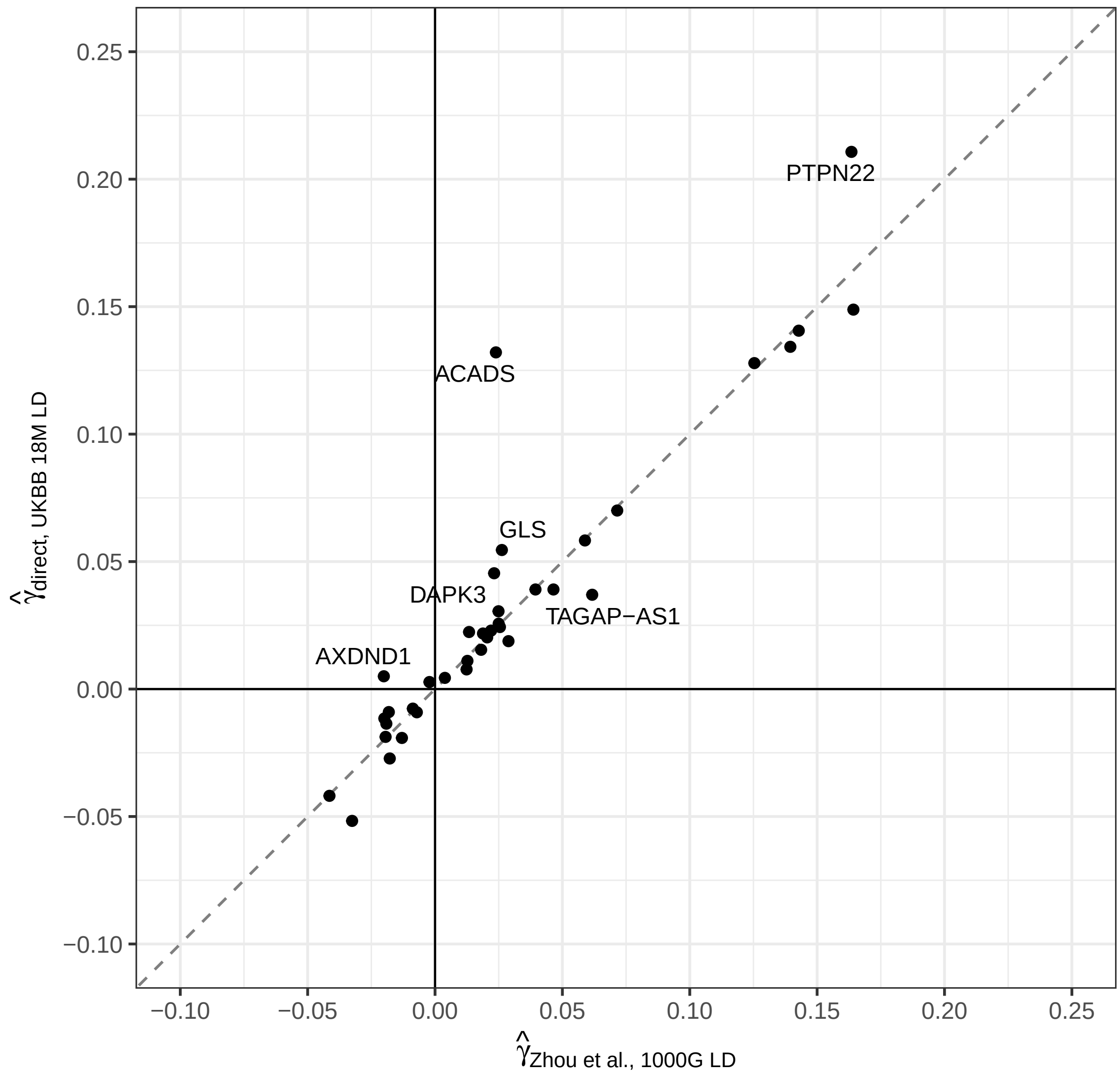

Figure 5: Both axes show directly computed $\hat{\gamma}$ from individual-level logistic regression of T1D on the scalar instrument score (per SD, log-odds). x-axis: Zhou et al. Analysis 1 using 1000 Genomes EUR LD weights. y-axis: new analysis using UKBB 18M EUR LD weights (Analysis 2). Both analyses use the same SDRNT1BIO/GS individual-level genotypes; they differ only in the LD reference panel used to derive instrument weights $\hat{\beta}_m$. Loci matched by chromosomal overlap. The six largest outliers from the identity line are labelled. Grey dashed line: identity ($y = x$).

11 Comparison of $\hat{\gamma}$: direct individual-level regression vs LD approximate

The direct method (Analysis 2, x-axis, ground truth) regresses T1D on individual instrument scores using UKBB EUR LD weights. The LD approximate method (Analysis 3, y-axis) estimates $\hat{\gamma}$ from the same per-SNP SNP-outcome coefficients via the projection formula in Section~2. Both use identical instrument weights $\hat{\beta}_m$. Attenuation of the LD approximate estimate reflects mismatch between the UKBB reference panel LD and the true LD in the GS target dataset. The Pearson correlation is $r = 1$.

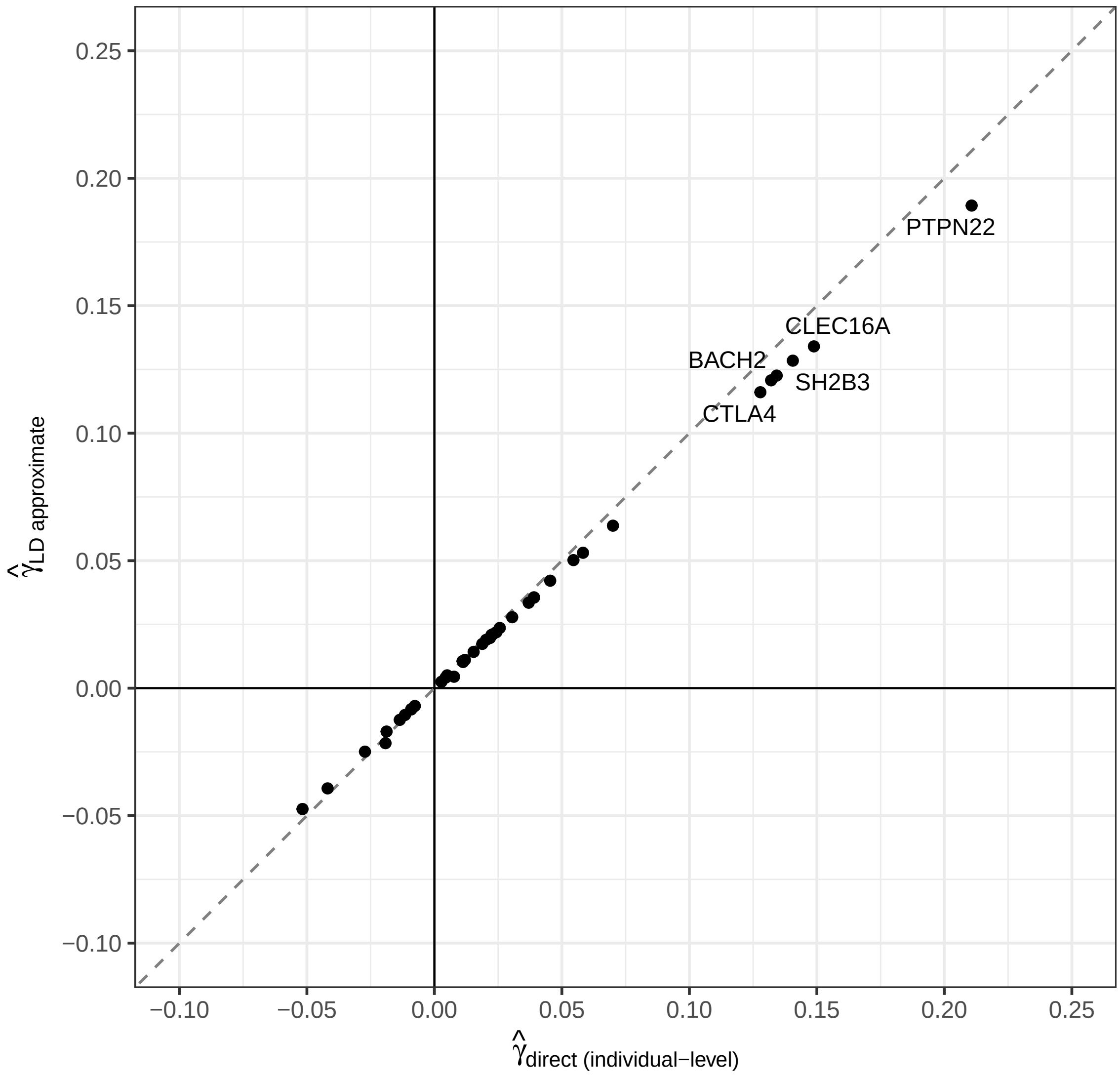

Figure 6: Direct individual-level regression $\hat{\gamma}$ (x-axis, ground truth) vs LD approximate $\hat{\gamma}$ (y-axis). Both per SD of scalar instrument, log-odds units; same UKBB 18M EUR LD weights used for both. Pearson $r = 1$. The six largest outliers from the identity line are labelled. Grey dashed: identity ($y = x$).

MR-Hevo posterior results not yet available. Run `run_mrhevo_pdcd1_t1d.R` *to generate* `mrhevo_pdcd1_t1d_fit.rds`*.*